

MATH2315 Linear Algebra I

Midterm Exam

October 2026

This exam has two questions. You must complete both questions to achieve full marks. This exam is worth 30 marks.

Duration: 90 minutes

1. Consider the collection $V = (S \times S, \oplus, \otimes)$, where

$$\begin{aligned} S &= \{x \mid x \in \mathbb{R}, x > 0\}, \\ (a, b) \oplus (c, d) &= (ac, bd) \text{ (coordinate multiplication),} \\ \lambda \otimes (a, b) &= (a^\lambda, b^\lambda) \text{ (coordinate exponentiation),} \end{aligned}$$

for all $v, w \in S \times S$, and $\lambda \in \mathbb{R}$. It can be shown V is a vector space where $\oplus$ plays the role of addition of vectors and $\otimes$ plays the role of scalar multiplication.

- (a) Find the identities of the operations $\otimes$ and $\oplus$. (2)
(i.e. find $\hat{e} \in V$ and $e \in \mathbb{R}$ such that $\hat{e} \oplus v = v$ and $e \otimes v = v$ for all $v \in V$.)
- (b) For any $v = (a, b) \in V$ and $\lambda \in \mathbb{R}$ find the additive ($\oplus$) inverse and the scaling ($\otimes$) inverse. (2)
(i.e. for all $v \in V$ and $\lambda \in \mathbb{R}$ find $\hat{w} \in V$ and $\rho \in \mathbb{R}$ such that $v \oplus w = \hat{e}$ and $(\rho\lambda) \otimes v = v$.)
- (c) Define *linear independence* of a set of vectors. (2)
- (d) Show that (4)

$$B = \{(1, 2), (2, 1)\}$$

is a linearly independent set in V and $\text{Span}(B) = V$. Hence B is a basis for V .

- (e) Represent the vectors $(3, 4)$ and $(\frac{1}{\pi}, 2)$ as linear combinations of B . (3)
- (f) Find the dual basis for B . (2)

2. Consider the vector space $V = (\mathbb{R}^3, +, \cdot)$, a basis

$$A = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

in V , and the map

$$L : V \rightarrow V : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} z \\ 2y - z \\ x \end{pmatrix}.$$

- (a) Prove that L is a linear map. (3)
- (b) State the *Rank-Nullity theorem*. (2)
- (c) Find the kernel and image of L ($\ker(L)$ and $\text{Im}(L)$) and their respective dimensions. (5)
- (d) Determine a basis of the image of L . (2)
- (e) Write L_A , the matrix form of L in terms of the basis A . Check your answer by comparing $(Lv)_A$ and $L_A v_A$, where (3)

$$v = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$